

Jane Simko, PhD

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PROFESSIONAL SUMMARY

ML/Computational Neuroscientist with 12+ years in applied research and building data pipelines and 5+ years training, deploying models, and delivering **real-time** ML systems from scratch. **Expert Python** programmer with hands-on experience across the **full ML lifecycle**: feature engineering, model training, evaluation, and deployment. Track record of producing models with measurable impact, integrating LLM-powered workflows, and **collaborating** across multiple interdisciplinary teams. Background in **time-series/biophysical** modeling and **signal processing** relevant to data science problems.

EXPERIENCE

Columbia University, Department of Neurology | New York, NY
Postdoctoral Research Fellow/Research Scientist | 3/2021 - Present

- Developed an unsupervised NLP pipeline on NIH All of Us survey data to identify social determinants distinguishing opioid use disorder cases from controls; embeddings via Hugging Face sentence-transformers.
- Fine-tuned BioBERT with LoRA/PEFT on clinical free-text and structured EHR in a multimodal late-fusion architecture for substance use disorder risk stratification.
- Built and maintained end-to-end ML pipelines for ingestion, preprocessing, feature engineering, and modeling of large-scale multimodal datasets.
- Designed, trained, and deployed a 1D CNN in PyTorch for real-time classification (95% accuracy), integrated with custom hardware for closed-loop inference at sub-second latency.
- Developed automated data quality checks, serialized preprocessing scalars for consistent inference-time normalization, and modular pipeline components reused across multiple projects.
- Mentored 6 researchers; secured 2 federal grants; authored 9+ publications.

École Polytechnique Fédérale de Lausanne | Lausanne, Switzerland
PhD Researcher | 1/2016 - 1/2021

- Developed computational models and large data pipelines to simulate large-scale neuronal population dynamics and biophysics using NEURON and high-performance computing clusters.
- Completed multiple full-cycle research projects from data collection to publication (*Cell Reports*, *J. Physiology*).

Penumbra Inc. | Alameda, CA
Neurotech Clinical Research Associate/Trial Coordinator | 8/2011 - 5/2013

- Led coordination of multi-site neurotechnology clinical trials for Class III medical devices, ensuring data integrity, regulatory compliance, and cross-functional team alignment.
- Contributed to protocol development and operational planning for early-stage devices in a growing med-tech environment.

UC Berkeley, Department of Electrical Engineering and CS | Berkeley, CA
Research Intern | 1/2011 - 6/2011

- Diagnosed and resolved technical issues during live experimental sessions; developed and debugged scripts for data acquisition.

EDUCATION

École Polytechnique Fédérale de Lausanne **PhD in Neuroscience**
1/2016 - 1/2021 | Lausanne, Switzerland

Université de Lausanne **MSc in Neuroscience**
8/2013 - 1/2015 | Lausanne, Switzerland

UC Berkeley **BA in Integrative Biology**
8/2007 - 5/2011 | Berkeley, CA

SKILLS

- **Programming:** Python (expert), R, SQL, C++, MATLAB
- **ML & Data:** PyTorch, TensorFlow, Scikit-learn, NumPy, SciPy, Pandas, MLflow, Git, NEURON
- **Signal & Data:** time-series processing, feature engineering, distributed data processing, embedding-based representations
- **AI/LLM:** Hugging Face Transformers, agentic AI workflows (Claude Code), LLM integration, GCP
- **Experimentation:** Experimental design, statistical validation, A/B testing methodology
- **Languages:** English (native), French (proficient)

SELECTED PUBLICATIONS

- 9 peer-reviewed publications + 4 preprints (Nature Communications, Cell Reports, J. Physiology, Cell Chem Bio).
- Full list via ORCID: 0000-0002-3632-4176.

AWARDS

- **BRAIN Initiative Individual Postdoctoral Fellowship (F32)**
9/2022 - Present
- **Swiss National Science Foundation Early Postdoc.Mobility Fellowship**
3/2021 - 9/2022
- **EPFL-Stanford Exchange Program Fellowship**
12/2018
- **Allen Institute Summer Workshop Travel Grant**
5/2017

LEADERSHIP & OUTREACH

- President, Young Swiss Society for Neuroscience, 2019
- 1st Swiss Early-Career Researchers Symposium, conference organizer, 2019
- Lead Organizer, Pint of Science, 2019
- Exposure Science Film Hackathon, 2017